Experiment :- Study of LAN Topology, network devices and address configuration.

Objective :-

1) To study the LAN topology used in college compus.
2) To identify networking device & understand their function.
3) To analyze wired & wireless communication method used in the network.
4) To measure speed of network for wired & wireless connections.
5) To obtained MAC, IP Address & subnet mask.

→ Problem statement - 1

- Draw & identify LAN topology of college compus & clearly label all components such as rooter, switches. & etc

Solution:- The college campus network observed this study consists of:

- Layer-3 core switches in the server Room.
- managed by POE + switches
- wireless Access point (APs)
- Ethernet port in classroom & laboratories

→ Network Topology

- Hybrid Hierarchical topology which combines.
  1) Tree Topology
  2) Extended Star Topology
  3) Ring Topology (server room)

→ The network hierarchical layer :-

(i) Core layer
(ii) Distribution layer
(iii) Access layer

(i) Core Layer :-

- core layer contains server Room and server room Now Firewall, Rounter, 2-L3 + switches, patch Rack, Server, Inverter etc

(ii) Distribution layer :-

- Distribution layer contains 4 - switches
- The 4 - switches are →

(i) main Build → R7 (Room no - 213, 2nd floor)

(ii) main Build → R1 (Room no - 301, 3rd floor)

(iii) main Build → R5 (Room no - 113, 1st floor)

(iv) main Build → R2 (Room no - 201, 2nd floor)

* This 4 - switches contains →

(1) 18 port Gigabit Easy Smart switch with 16 - Port

TL - SG2218MPE

(2) SG3428XMP → 24 port Gigabit & 4 - Port 10GE SFP+, L2+ managed switches with 24 Port POE+.

→ Now let's discuss these 2 above complements

① TP-Link TL-SG1218MPE

- Linkely used for: CCTV camera, wifi access point etc
- 18 port Gigabit "Easy Smart" switch.
- 16 Port Support POE+

② TP-Link Omada SG3428XMP

- Higher-end managed Layer 2+ switch.
- 24 POE+ port, uplinks, 10G SFP.
- Usually acts as a main distribution for each floor.

→ Access Layer :—

- Access layer contain 3-swtiches i.e

1) main Build R1 (Room no-301, 3rd Floor)
2) main Build R4 (Room no-05, Ground floor)
3) Main Build R5 (Room no-18, Ground Floor)

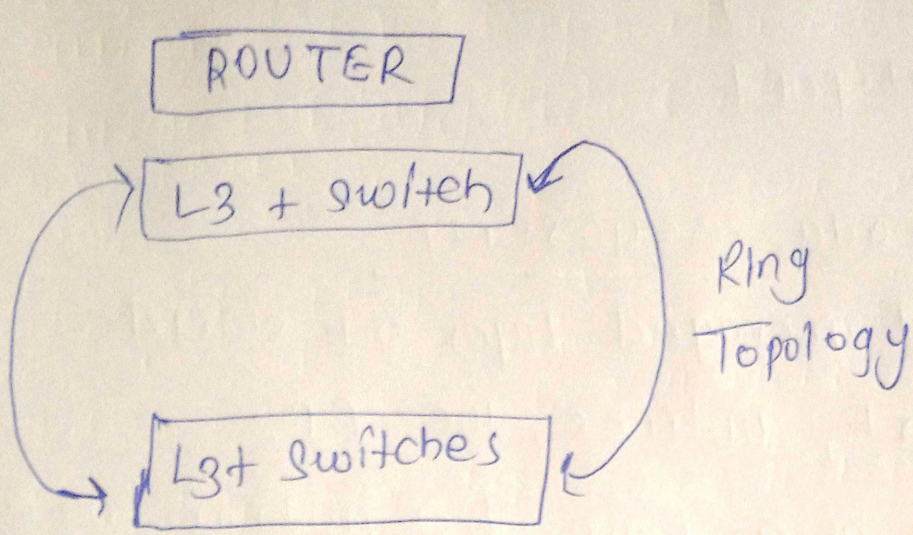
* Every switches acts as access layer which helps to provide network connectivity to all class Room lab, Faculity Room, etc.

* Access layer directly connect :—

(1) Ethernet port
(2) Wireless AP
(3) Lab system

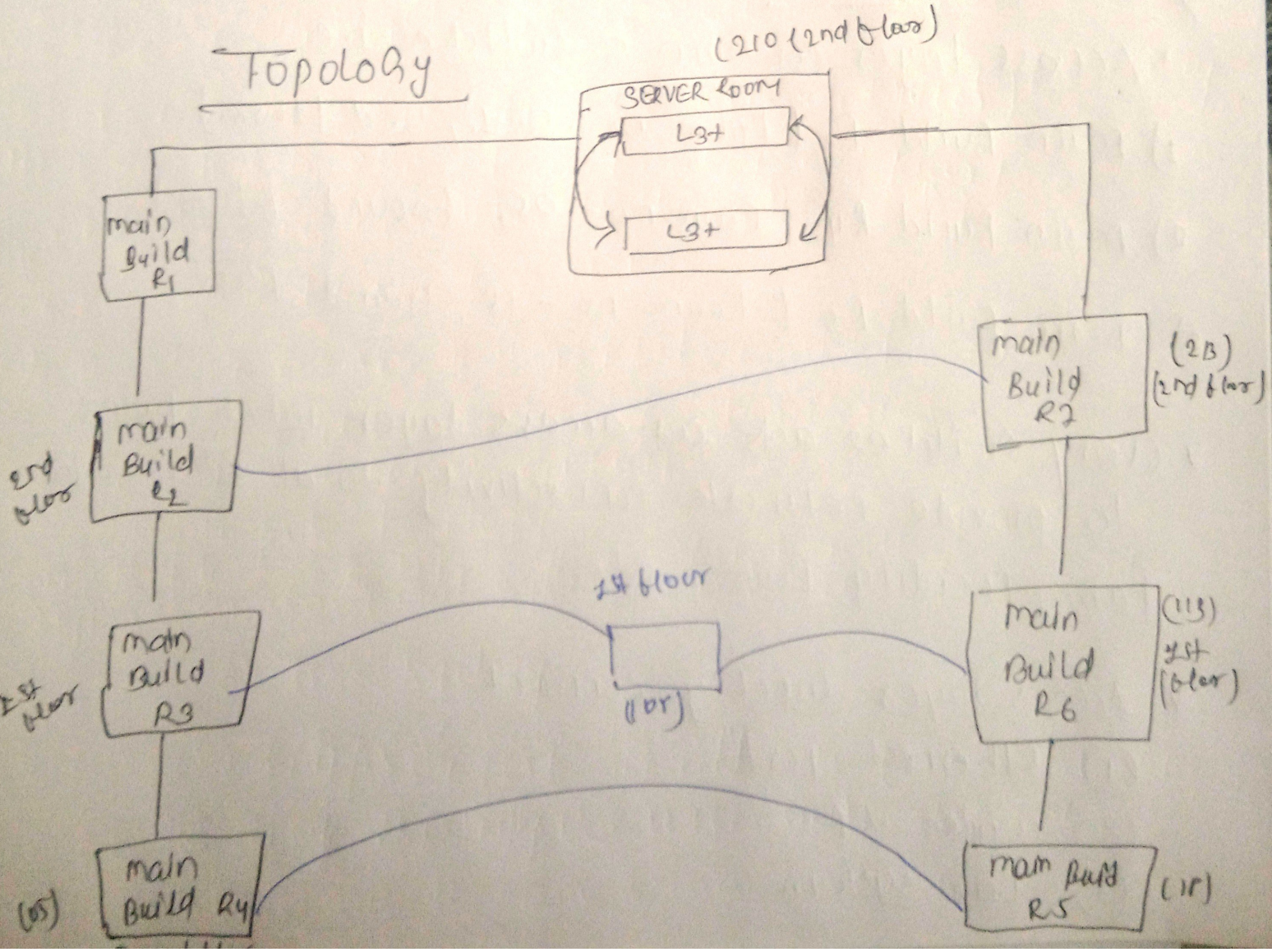
(9) Outdoor wireless AP (ex - 018)

→ SERVER Room Architecture / Topology



L2 + switches

Topology

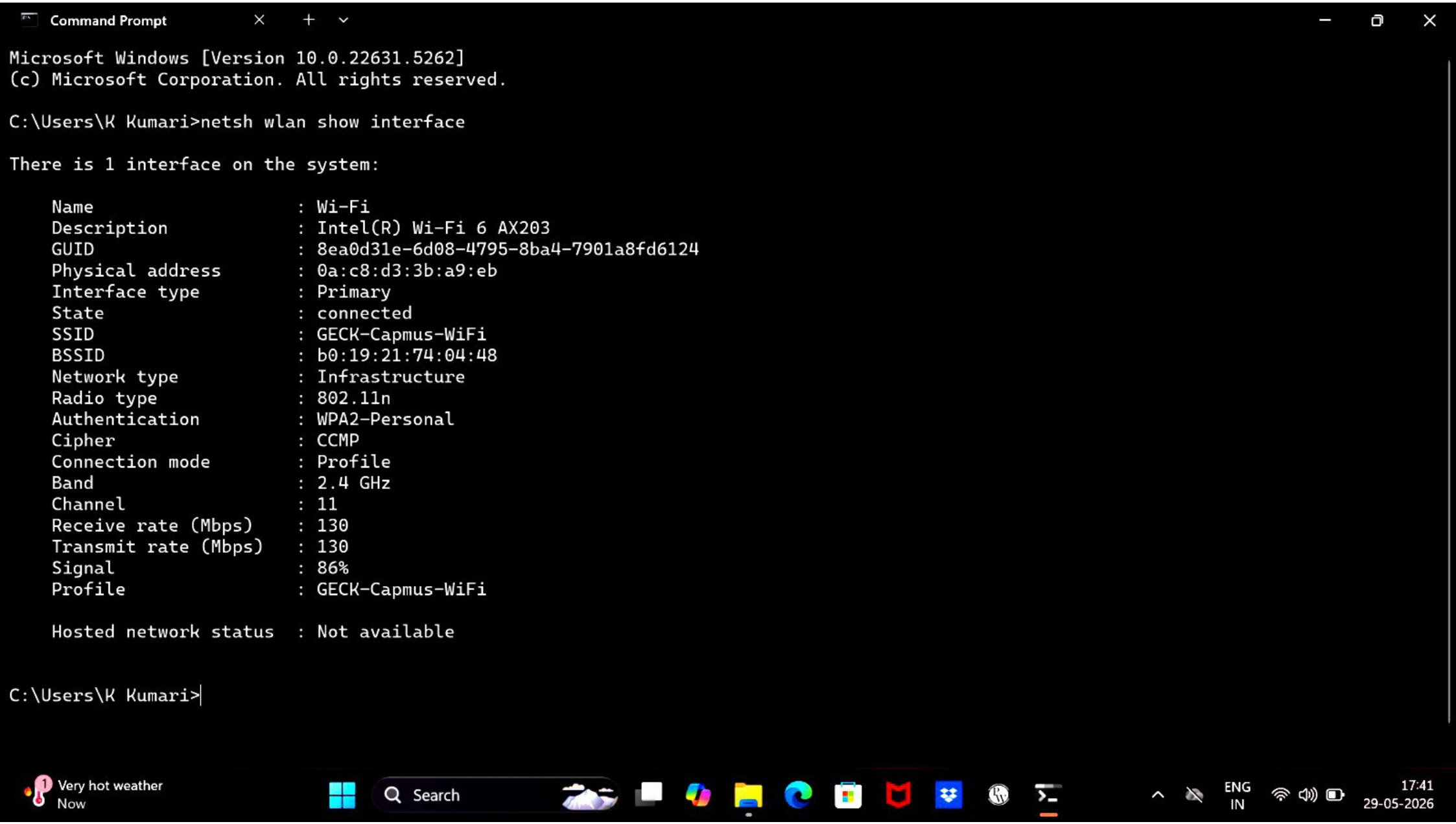


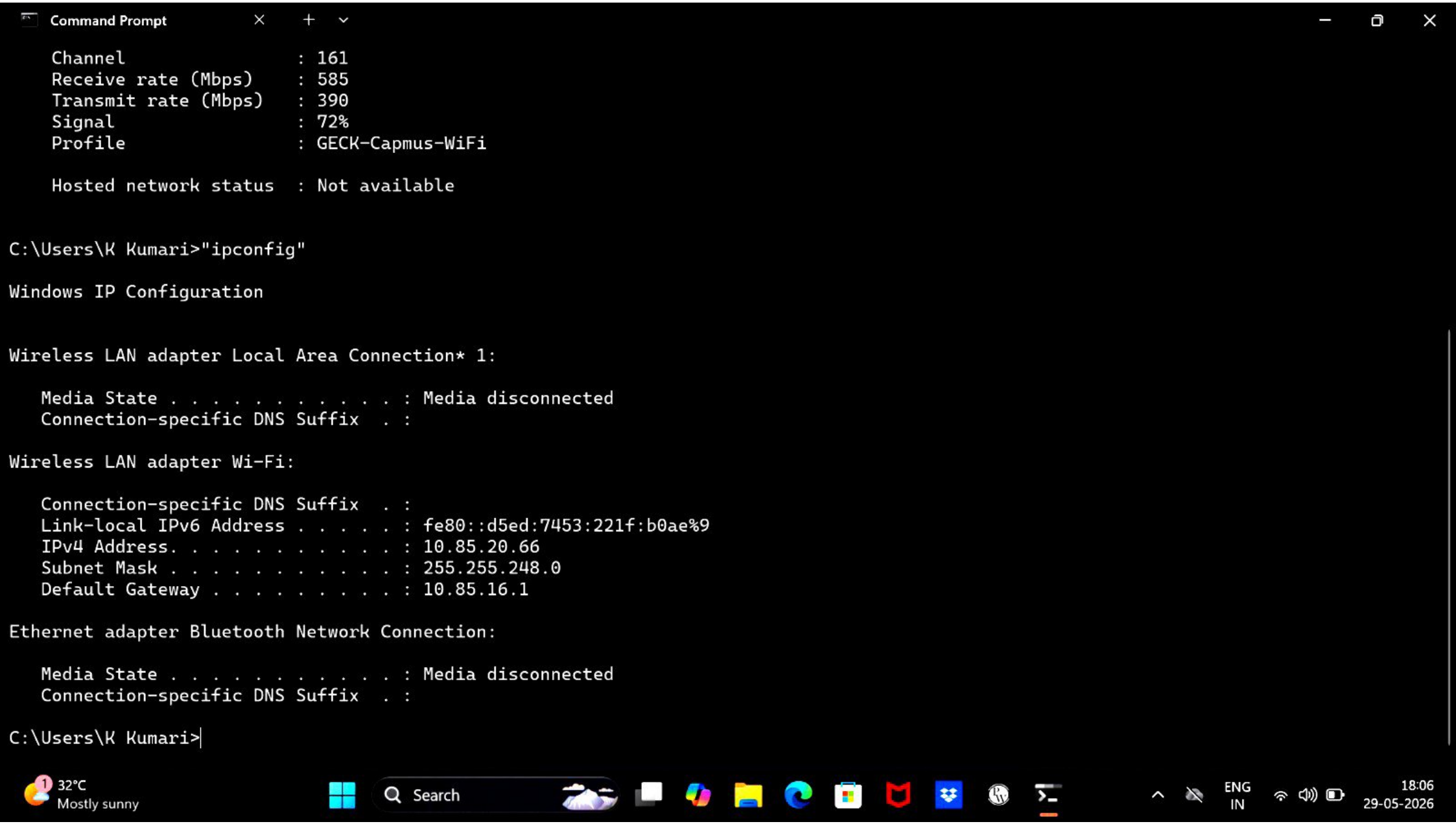
→ R_1, R_7, R_2, R_6 → Distribution layer

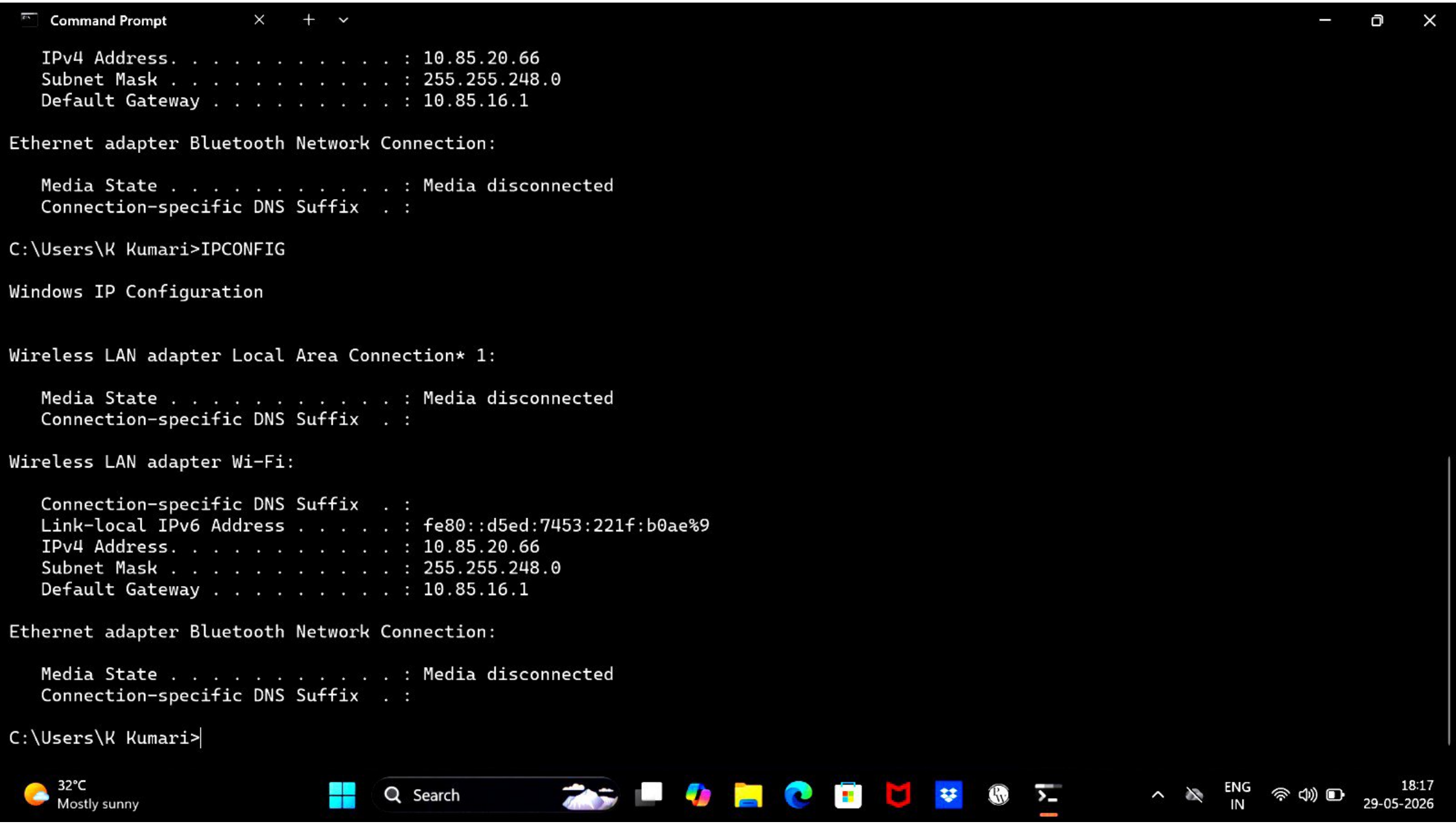
→ Server Room → Core layer

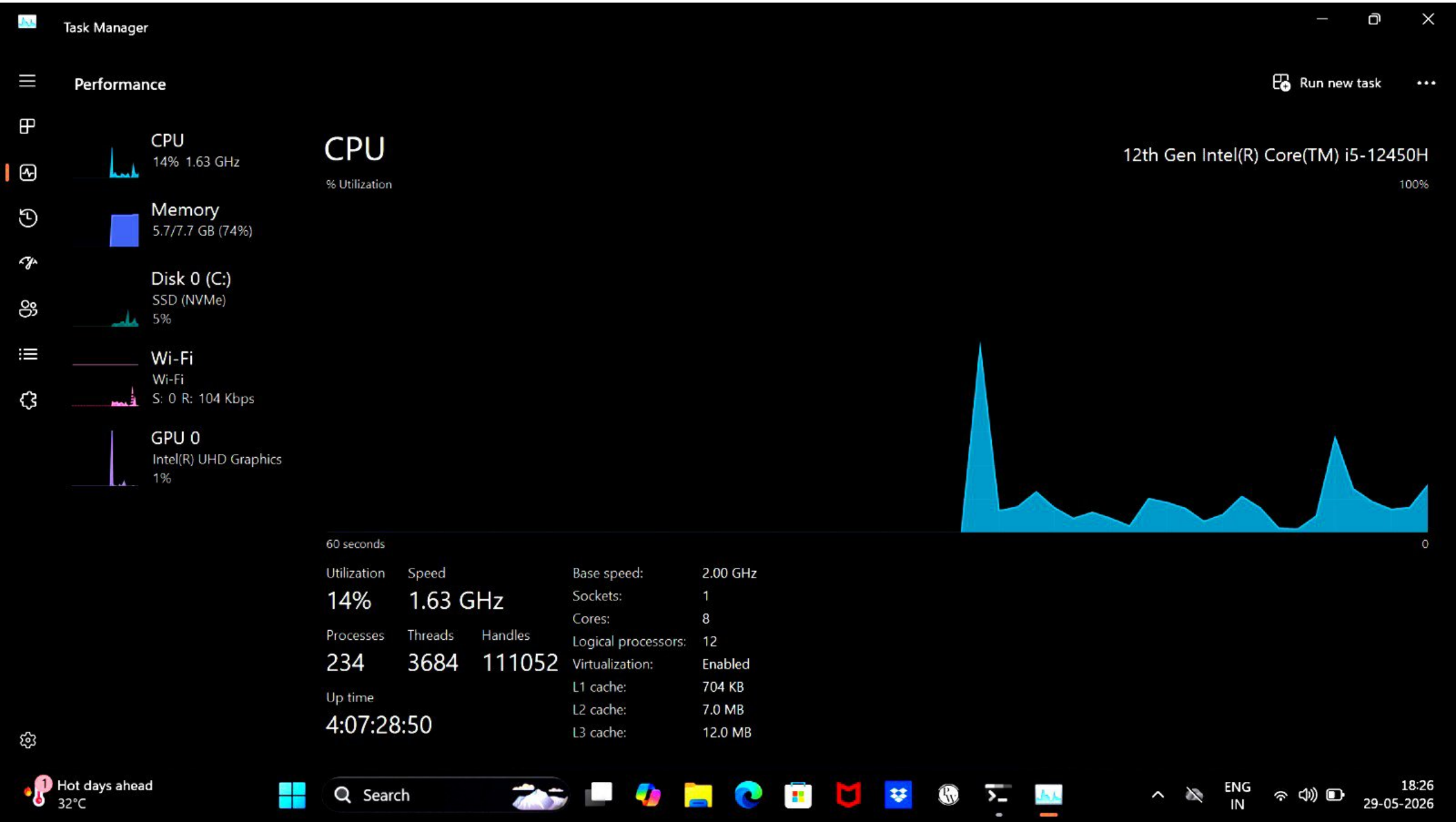
→ R_1, R_2, R_3, R_4, R_5, R_6, R_7 → Access layer

- ACCESS LAYER SWITCH (3rd Floor - 301)


Command Prompt
Microsoft Windows [Version 10.0.22631.5262]
(c) Microsoft Corporation. All rights reserved.
C:\Users\K Kumari>netsh wlan show interface
There is 1 interface on the system:
Name : Wi-Fi
Description : Intel(R) Wi-Fi 6 AX203
GUID : 8ea0d31e-6d08-4795-8ba4-7901a8fd6124
Physical address : 0a:c8:d3:3b:a9:eb
Interface type : Primary
State : connected
SSID : GECK-Capmus-WiFi
BSSID : b0:19:21:74:04:48
Network type : Infrastructure
Radio type : 802.11n
Authentication : WPA2-Personal
Cipher : CCMP
Connection mode : Profile
Band : 2.4 GHz
Channel : 11
Receive rate (Mbps) : 130
Transmit rate (Mbps) : 130
Signal : 86%
Profile : GECK-Capmus-WiFi
Hosted network status : Not available
C:\Users\K Kumari>
Very hot weather
Now
Search
ENG
IN
17:41
29-05-2026


Command Prompt
Channel : 161
Receive rate (Mbps) : 585
Transmit rate (Mbps) : 390
Signal : 72%
Profile : GECK-Capmus-WiFi
Hosted network status : Not available
C:\Users\K Kumari>"ipconfig"
Windows IP Configuration
Wireless LAN adapter Local Area Connection* 1:
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Wireless LAN adapter Wi-Fi:
Connection-specific DNS Suffix . :
Link-local IPv6 Address . . . . . : fe80::d5ed:7453:221f:b0ae%9
IPv4 Address. . . . . : 10.85.20.66
Subnet Mask . . . . . : 255.255.248.0
Default Gateway . . . . . : 10.85.16.1
Ethernet adapter Bluetooth Network Connection:
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
C:\Users\K Kumari>
32°C
Mostly sunny
Search
ENG
IN
18:06
29-05-2026


Command Prompt
IPv4 Address. . . . . : 10.85.20.66
Subnet Mask . . . . . : 255.255.248.0
Default Gateway . . . . . : 10.85.16.1
Ethernet adapter Bluetooth Network Connection:
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
C:\Users\K Kumari>IPCONFIG
Windows IP Configuration
Wireless LAN adapter Local Area Connection* 1:
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Wireless LAN adapter Wi-Fi:
Connection-specific DNS Suffix . :
Link-local IPv6 Address . . . . . : fe80::d5ed:7453:221f:b0ae%9
IPv4 Address. . . . . : 10.85.20.66
Subnet Mask . . . . . : 255.255.248.0
Default Gateway . . . . . : 10.85.16.1
Ethernet adapter Bluetooth Network Connection:
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
C:\Users\K Kumari>
32°C
Mostly sunny
Search
ENG
IN
18:17
29-05-2026


Task Manager
Performance
Run new task
CPU
14% 1.63 GHz
Memory
5.7/7.7 GB (74%)
Disk 0 (C:)
SSD (NVMe)
5%
Wi-Fi
Wi-Fi
S: 0 R: 104 Kbps
GPU 0
Intel(R) UHD Graphics
1%
CPU
% Utilization
12th Gen Intel(R) Core(TM) i5-12450H
100%
60 seconds
0
Utilization
14%
Speed
1.63 GHz
Processes
234
Threads
3684
Handles
111052
Up time
4:07:28:50
Base speed: 2.00 GHz
Sockets: 1
Cores: 8
Logical processors: 12
Virtualization: Enabled
L1 cache: 704 KB
L2 cache: 7.0 MB
L3 cache: 12.0 MB
Hot days ahead
32°C
Search
ENG
IN
18:26
29-05-2026

→ Difference b/w Wired & wireless speed

Wired Network	Wireless Network
• Faster communication	• Slower communication
• Stable connection	Affected by distance / interference
• Low latency	• Higher latency
• Dedicated physical medium.	• Shared radio medium

→ • Ethernet provides · Uses radio waves which are affected by inter ference, distance, obstacles & multiple connected users

Q Role of subnet mask in network?

• A subnet mask identifies the network portion & host portion of an IP address • It helps devices determine whether another device belongs to the same network or different network.

It is important for :-

(1) Network segment

(ii) Efficient routing

(iii) Communication b/w devices

OBSERVATION :—

During the study of the college network infrastructure :—

(1) A centralised server room was used as network backbone.

(2) Two L3 switches were connected in redundant cyclic structure.

(3) managed TP-Link POE+ switches were used in network cabinets

→ Advantage of Existing Network?

- Scalable architecture.
- Supports POE device
- Easy expansion of floors & classroom
- Redundant link for failover.

CONCLUSION :—

The observed campus LAN follow a Hybrid hierachial topology consist of tree, extended star, Ring. The network uses managed POE switches, wireless access point ethernet port & controlled server infrastructure to provide efficient wired & wireless communication, access the campus.